

# **METHODOLOGY REPORT: STUDY BUDDY**

Computer Science II

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## **INTRODUCTION**

**Study Buddy** is a productivity-based academic platform designed specifically for students. The system integrates task management, grade tracking, and quiz generation into a single centralized dashboard. Its goal is to reduce the fragmentation of academic systems and centralize academic monitoring.

This methodology report is intended to document the group's journey in developing the initial frameworks and core features necessary for the intended program, which is Study Buddy.

## **RESEARCH AND PLANNING PHASE**

After the presentation stage, the group entered the initial development stage. They analyzed common academic productivity challenges, including scattered systems and inefficient task monitoring. Sample grading systems, assessment weights, and system flow structures were collected to simulate realistic academic environments. A blueprint of the program's modular framework was created before coding.

## **METHODOLOGY DEVELOPMENT PHASE:**

The group followed an Incremental and Modular development approach to reduce the risk of recurring errors and allow them to obtain early user feedback. Each core feature (Task Manager, Grade Calculator, Quiz Generator) was developed and implemented individually by team members before any system integration.

## **SYSTEM BLUEPRINT**

Our system follows Client-Server Architecture. The server component is implemented using Python and Flask, handling all data processing, grade calculations, and quiz generation. The User Interface (UI)

utilizes HTML, CSS, and JavaScript to manage user interaction and content rendering. JSON files function as a lightweight tool for local data storage.

## CORE FEATURES IMPLEMENTATION

Once all the core features were developed separately, all of them were implemented into one program using Visual Studio Code (VSCode).

1. Smart Task Manager [MARQUEZ W]: Enables subject-based task entry with deadlines stored in JSON format. Initially, it included smart task categorization based on alertness levels; however, this feature was later removed to streamline functionality and improve usability.
2. Grade Calculator [MORGA G. AND COSTO Z.]: Implements weighted average computation using score, maximum score, and weight inputs for both PSHS and DepEd system grading.
3. Quiz Generator [DUROY P]: Utilizes structured JSON question banks and random sampling for quiz creation.

Each module underwent unit-level testing before integration. Manual test cases validated weighted grade accuracy, data persistence, and random quiz generation reliability. Error-handling mechanisms were implemented for invalid inputs and missing data scenarios. However, there were still errors despite these precautions.

As of the **4th quarter**, there was a significant need for system improvements. Major changes were implemented, including the adoption of the **CRUD (Create, Read, Update, Delete)** standard across the program to enhance data management and consistency. This required **extensive revisions** in each module, marking a substantial overhaul of the system after the first project presentation.

Additionally, during the 4th quarter, **MARQUEZ W.** took primary responsibility for handling and managing task-related features and updates within the system.

## AREAS FOR IMPROVEMENT

Time management, coordination, and documentation updates were identified as development challenges. Future improvements include database migration (e.g., Liquibase, cloud services), user authentication, enhanced UI responsiveness, and automated testing. This will soon be updated once the 4<sup>th</sup> Quarter semester starts.

**4<sup>TH</sup> QUARTER:** As of the final quarter, these developmental challenges persisted; however, they did not significantly hinder the overall progress of the application. Database migration was not implemented due to the limited scope of the project. Nevertheless, it is recognized as a potential area for improvement and may be considered for future updates and system enhancements.

## CONCLUSION

The Study Buddy program has undergone **multiple system modifications** and remains **under continuous development**. Despite this, the current version of the application demonstrates its intended functionality and potential. For the purposes of the final presentation, the system is considered **complete**; however, further improvements and enhancements are anticipated in future iterations.

## References

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